ECA5801-IOE AND AI DRIVEN SYSTEMS

EXPERIMENT 5.A

SIMULATION

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A.Simulation and Performance Evaluation of People–Process–Data–Things IoE Framework using Edge/Fog-to-Cloud Pipeline with MQTT, Node-RED.

CODE:

```
# ============================================================
# EXPERIMENT A
# P-P-D-T IoE Framework using Edge/Fog-to-Cloud
# MQTT and Node-RED Simulation
# ============================================================
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
# --------------------------------
# 1. Simulation Parameters
# --------------------------------

np.random.seed(42)

layers = ["Edge", "Fog", "Cloud"]

# Number of IoT data packets
num_packets = 1000

# Packet size in KB
packet_size = 2
```

```
# Simulation time in seconds
simulation_time = 10

# ------------------------------
# 2. Parameters for Edge, Fog, Cloud
# ------------------------------

throughput_capacity = {
    "Edge": 95,
    "Fog": 70,
    "Cloud": 50
}

base_latency = {
    "Edge": 10,
    "Fog": 25,
    "Cloud": 60
}

packet_loss = {
    "Edge": 0.02,
    "Fog": 0.04,
    "Cloud": 0.06
}

# ------------------------------
# 3. Simulation
# ------------------------------

results = []
```

```
for layer in layers:

    # Packet delivery
    delivered_packets = int(
        num_packets * (1 - packet_loss[layer])
    )

    # Throughput in Mbps
    total_data_bits = delivered_packets * packet_size * 8 * 1024

    throughput = total_data_bits / simulation_time / 1e6

    # Limit according to layer capacity
    throughput = min(
        throughput,
        throughput_capacity[layer]
    )

    # Latency in milliseconds
    latency = base_latency[layer] + np.random.uniform(2, 8)

    results.append([
        layer,
        delivered_packets,
        throughput,
        latency
    ])

# --------------------------------
# 4. Create Result Table
# --------------------------------
```

```
df = pd.DataFrame(
    results,
    columns=[
        "Layer",
        "Packets Delivered",
        "Throughput (Mbps)",
        "Latency (ms)"
    ]
)

df["Throughput (Mbps)"] = df["Throughput (Mbps)"].round(2)
df["Latency (ms)"] = df["Latency (ms)"].round(2)

print("\n========== P-P-D-T IoE SIMULATION RESULTS ==========\n")
display(df)

# ------------------------------
# 5. Throughput Graph
# ------------------------------

plt.figure(figsize=(5, 3))

plt.bar(
    df["Layer"],
    df["Throughput (Mbps)"]
)

plt.title("Throughput: Edge vs Fog vs Cloud")
plt.xlabel("IoE Processing Layer")
plt.ylabel("Throughput (Mbps)")
```

```
plt.grid(axis="y", alpha=0.3)

plt.show()

# ------------------------------
# 6. Latency Graph
# ------------------------------

plt.figure(figsize=(5, 3))

plt.bar(
    df["Layer"],
    df["Latency (ms)"]
)

plt.title("Latency: Edge vs Fog vs Cloud")
plt.xlabel("IoE Processing Layer")
plt.ylabel("Latency (ms)")
plt.grid(axis="y", alpha=0.3)

plt.show()

# ------------------------------
# 7. 5G vs 6G Comparison
# ------------------------------

network = ["5G", "6G"]

network_throughput = [1000, 5000]
network_latency = [10, 1]
```

```
comparison = pd.DataFrame({
    "Network": network,
    "Throughput (Mbps)": network_throughput,
    "Latency (ms)": network_latency
})

print("\n========== 5G vs 6G COMPARISON ==========\n")
display(comparison)

# --------------------------------
# 8. 5G vs 6G Graph
# --------------------------------

fig, ax = plt.subplots(1, 2, figsize=(8, 3))

ax[0].bar(
    comparison["Network"],
    comparison["Throughput (Mbps)"]
)

ax[0].set_title("5G vs 6G Throughput")
ax[0].set_ylabel("Mbps")

ax[1].bar(
    comparison["Network"],
    comparison["Latency (ms)"]
)

ax[1].set_title("5G vs 6G Latency")
ax[1].set_ylabel("ms")
```

```
plt.tight_layout()
plt.show()

# ------------------------------
# 9. Conclusion
# ------------------------------

print("\n========== CONCLUSION ==========")

best_throughput = df.loc[
    df["Throughput (Mbps)"].idxmax(),
    "Layer"
]

lowest_latency = df.loc[
    df["Latency (ms)"].idxmin(),
    "Layer"
]

print("• Edge provides the highest throughput in the simulated IoE pipeline.")
print("• Edge provides the lowest latency because processing occurs closer to the device.")
print("• Fog provides an intermediate processing layer between Edge and Cloud.")
print("• Cloud introduces higher latency because data travels farther.")
print("• 6G provides higher throughput and lower latency than 5G in the comparison model.")
```

OUTPUT:

```
========== RFID/NFC IDENTIFICATION TABLE ==========
```

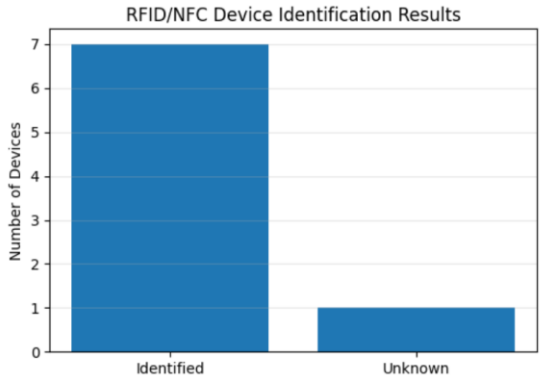
	Device ID	Technology	Device Type	Identification	MQTT Message
0	RFID_101	RFID	Student ID	Identified	RFID_101:IDENTIFIED
1	NFC_102	NFC	Mobile Phone	Identified	NFC_102:IDENTIFIED
2	RFID_103	RFID	Library Book	Identified	RFID_103:IDENTIFIED
3	NFC_104	NFC	Access Card	Identified	NFC_104:IDENTIFIED
4	RFID_105	RFID	Equipment	Identified	RFID_105:IDENTIFIED
5	NFC_106	NFC	Mobile Phone	Identified	NFC_106:IDENTIFIED
6	RFID_107	RFID	Student ID	Identified	RFID_107:IDENTIFIED
7	NFC_108	NFC	Access Card	Unknown	NFC_108:REJECTED

```
========== SIMULATED MQTT MESSAGES ==========
MQTT publish: RFID_101:IDENTIFIED
MQTT publish: NFC_102:IDENTIFIED
MQTT publish: RFID_103:IDENTIFIED
MQTT publish: NFC_104:IDENTIFIED
MQTT publish: RFID_105:IDENTIFIED
MQTT publish: NFC_106:IDENTIFIED
MQTT publish: RFID_107:IDENTIFIED
MQTT publish: NFC_108:REJECTED

========== NODE-RED FLOW ==========
RFID/NFC Reader -> MQTT Input -> Node-RED Processing
-> Device ID Check -> Dashboard Update

========== CLOUD DASHBOARD ==========
```

	Metric	Value
0	Devices scanned	8.0
1	Successfully identified	7.0
2	Unknown devices	1.0
3	Identification success rate (%)	87.5




```
========== CONCLUSION ==========
Devices scanned: 8
Successfully identified: 7
Unknown devices: 1
Four layers: Perception -> Network -> Middleware -> Application
Data flow: RFID/NFC -> MQTT -> Node-RED -> Cloud Dashboard
Note: This is a simulated workflow, not a live MQTT connection.
```

EXPERIMENT 5B

b.Benchmarking of 5G/6G Communication for IoE Applications based on Throughput, Latency, Reliability and Energy Efficiency

CODE:

```
# ============================================================

# EXPERIMENT B

# Benchmarking of 5G/6G Communication for IoE Applications

# ============================================================


import numpy as np

import pandas as pd

import matplotlib.pyplot as plt


# ------------------------------

# 1. Network Parameters

# ------------------------------


networks = ["5G", "6G"]


# Throughput in Mbps

throughput = [1000, 5000]


# Latency in milliseconds

latency = [10, 1]


# Reliability in percentage

reliability = [99.0, 99.99]


# Energy efficiency
```

```
# Higher value = better efficiency
energy_efficiency = [70, 90]

# ------------------------------
# 2. Create DataFrame
# ------------------------------

df = pd.DataFrame({
    "Network": networks,
    "Throughput (Mbps)": throughput,
    "Latency (ms)": latency,
    "Reliability (%)": reliability,
    "Energy Efficiency (%)": energy_efficiency
})

print("\n========== 5G vs 6G IoE BENCHMARK ==========\n")

display(df)

# ------------------------------
# 3. Throughput
# ------------------------------

plt.figure(figsize=(5, 3))

plt.bar(
    df["Network"],
    df["Throughput (Mbps)"]
)
```

```
plt.title("5G vs 6G - Throughput")
plt.xlabel("Network")
plt.ylabel("Throughput (Mbps)")
plt.grid(axis="y", alpha=0.3)

plt.show()

# -------------------------------
# 4. Latency
# -------------------------------

plt.figure(figsize=(5, 3))

plt.bar(
    df["Network"],
    df["Latency (ms)"]
)

plt.title("5G vs 6G - Latency")
plt.xlabel("Network")
plt.ylabel("Latency (ms)")
plt.grid(axis="y", alpha=0.3)

plt.show()

# -------------------------------
# 5. Reliability
# -------------------------------
```

```
plt.figure(figsize=(5, 3))

plt.bar(
    df["Network"],
    df["Reliability (%)"]
)

plt.title("5G vs 6G - Reliability")
plt.xlabel("Network")
plt.ylabel("Reliability (%)")
plt.ylim(95, 100)

plt.grid(axis="y", alpha=0.3)

plt.show()

# ------------------------------
# 6. Energy Efficiency
# ------------------------------

plt.figure(figsize=(5, 3))

plt.bar(
    df["Network"],
    df["Energy Efficiency (%)"]
)

plt.title("5G vs 6G - Energy Efficiency")
```

```
plt.xlabel("Network")
plt.ylabel("Energy Efficiency (%)")

plt.grid(axis="y", alpha=0.3)

plt.show()

# ------------------------------
# 7. Combined Comparison
# ------------------------------

fig, ax = plt.subplots(2, 2, figsize=(8, 6))

ax[0, 0].bar(
    df["Network"],
    df["Throughput (Mbps)"]
)
ax[0, 0].set_title("Throughput")
ax[0, 0].set_ylabel("Mbps")

ax[0, 1].bar(
    df["Network"],
    df["Latency (ms)"]
)
ax[0, 1].set_title("Latency")
ax[0, 1].set_ylabel("ms")

ax[1, 0].bar(
    df["Network"],
```

```
    df["Reliability (%)"]
)
ax[1, 0].set_title("Reliability")
ax[1, 0].set_ylabel("%")

ax[1, 1].bar(
    df["Network"],
    df["Energy Efficiency (%)"]
)
ax[1, 1].set_title("Energy Efficiency")
ax[1, 1].set_ylabel("%")

plt.tight_layout()
plt.show()

# --------------------------------
# 8. Automatic Comparison
# --------------------------------

print("\n========== COMPARISON ==========\n")

if throughput[1] > throughput[0]:
    print("✓ 6G has higher throughput than 5G.")

if latency[1] < latency[0]:
    print("✓ 6G has lower latency than 5G.")

if reliability[1] > reliability[0]:
    print("✓ 6G has higher reliability than 5G.")
```

```
if energy_efficiency[1] > energy_efficiency[0]:
    print("✓ 6G has better energy efficiency than 5G.")


# ------------------------------
# 9. Conclusion
# ------------------------------


print("\n========== CONCLUSION ==========\n")

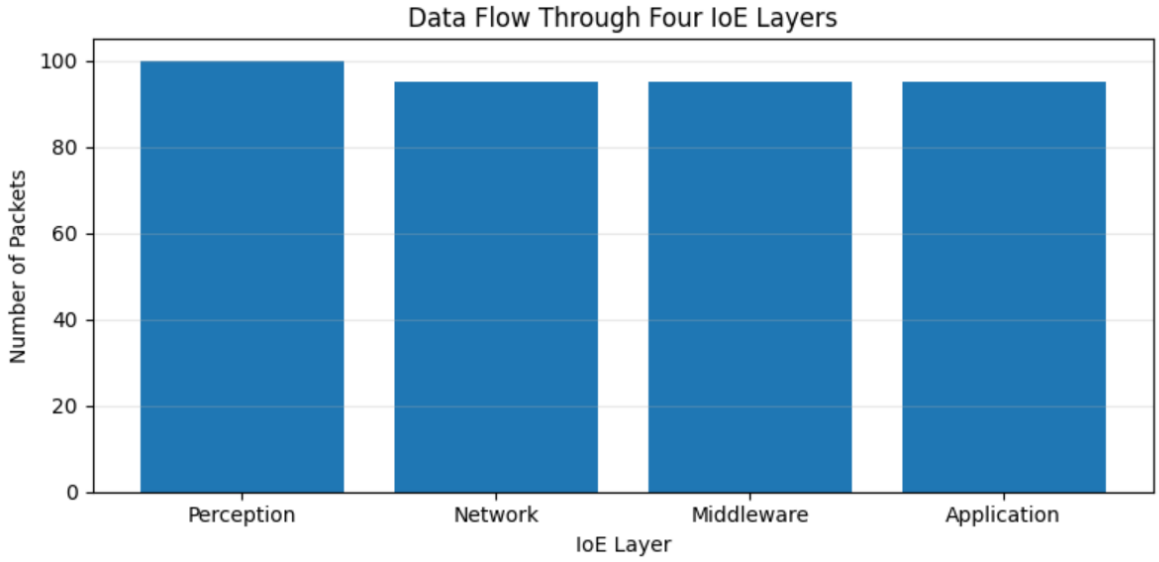

print("6G provides better simulated performance than 5G")
print("for IoE applications in terms of:")
print("• Higher throughput")
print("• Lower latency")
print("• Higher reliability")
print("• Better energy efficiency")
```

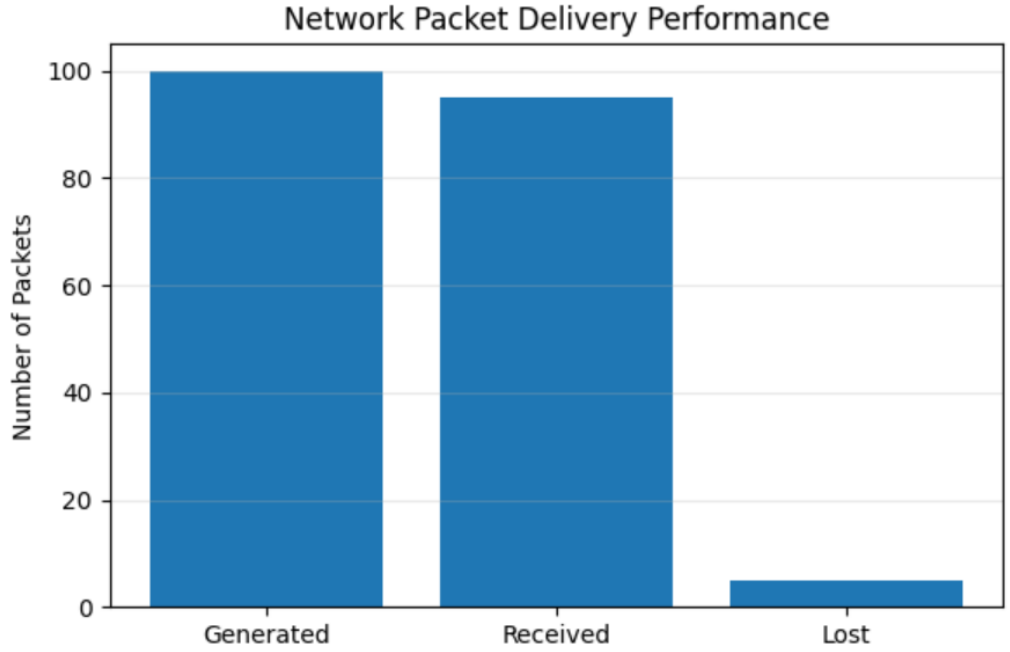
OUTPUT:

```
========== LAYER-WISE DATA FLOW ==========
```

	Layer	Input Packets	Output Packets	Function
0	Perception	0	100	Collect sensor readings
1	Network	100	95	Transfer packets
2	Middleware	95	95	Process received data
3	Application	95	95	Display dashboard information

```
========== SIMULATION STATUS ==========
Perception: 100 readings generated
Network: 95 packets received
Network: 5 packets lost
Middleware: 95 packets processed
Application: 95 packets displayed
```






```
========== PERFORMANCE SUMMARY ==========
```

	Performance Metric	Value
0	Total readings generated	100.0
1	Packets received	95.0
2	Packets lost	5.0
3	Packet delivery ratio (%)	95.0

```
========== CONCLUSION ==========
1. The Perception layer collects real-world data.
2. The Network layer transfers data between devices.
3. The Middleware layer processes the received data.
4. The Application layer presents information to users.
5. Packet delivery ratio = 95.00%
The four-layer IoE architecture was modelled and simulated.
```